

Occupancy detection

Programming assignment - Machine Learning

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1 Introduction

Domotics is the branch of automation dedicated to controlling home equipment. In particular, occupancy detection aims to develop automated systems capable of determining whether a room is occupied by monitoring environmental data.

The goal of this project is to train different machine learning models to classify a room as either occupied or empty based on multiple features. Two distinct models, a logistic regression model and a multilayer perceptron, are implemented, analyzed, and evaluated on a test set. Finally a precision and a recall-oriented version of each model are developed by tuning their classification threshold.

2 Data and features

The dataset comprises 8,143 records, each containing the following measurements:

1. Room temperature, measured in Celsius degrees.
2. Humidity percentage.
3. Light, measured in Lux.
4. CO₂ concentration, measured in parts per milion.
5. The humidity ratio, computed from the measured temperature and relative humidity.
6. The occupancy, labeled with 0 if the room is empty and 1 if it is full.

These data are divided into a training set of 6,943 records, a validation set of 400 records, and a test set of 800 records. To accelerate model convergence and improve weight interpretability, mean-variance normalization is applied to the dataset. This normalization process computes the mean μ_j and standard deviation σ_j for the feature x_j across all samples as follows:

$$\mu_j = \frac{1}{m} \sum_{i=0}^{m-1} \mathbf{x}_{ij} \quad (1)$$

$$\sigma_j = \sqrt{\frac{1}{m} \sum_{i=0}^{m-1} (x_{ij} - \mu_j)^2}. \quad (2)$$

The normalized feature $\bar{x}_j$ is then calculated as

$$\bar{x}_j = \frac{x_j - \mu_j}{\sigma_j}. \quad (3)$$

3 Logistic regression

The first model trained on the dataset is a logistic regression classifier. For a binary classification problem, logistic regression models the probability of a sample with feature vector $\mathbf{x}_i$ belonging to class 1 as

$$P(y = 1|\mathbf{x}) = \hat{p} = \sigma(\mathbf{w}^T \mathbf{x} + b) \quad (4)$$

where $\sigma(z)$ is the sigmoid function, defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}. \quad (5)$$

The optimal values of the weights $\mathbf{w}$ and bias b are found by minimizing the cross entropy loss function on the training dataset:

$$L(\mathbf{w}, b) = \frac{1}{m} \sum_{i=0}^{m-1} -y_i \log \hat{p}_i - (1 - y_i) \log(1 - \hat{p}_i). \quad (6)$$

This loss function is minimized numerically using gradient descent.

After training, the logistic regression model achieved an accuracy of 98.78% and 99.12% on the training and test set respectively, demonstrating good performance and generalization. Table 1 reports the final learned weights associated with each normalized feature. These results show that the single most predictive feature is the luminosity of the room followed by the concentration of CO₂. Both these features carry high positive weights, this is consistent with the expectation that human presence corresponds to active lighting and increased CO₂ emissions. Conversely, room temperature yields a negative weight despite an initial expectation of positive correlation with occupancy, this is likely due to the correlation between the features: since human activity increases all these features simultaneously, the human contribution to the temperature is already captured by the light and CO₂ weights and the model assigns a negative weight to temperature to compensate for environmental changes in temperature (high temperature without humans).

Feature	Weight
Temperature	-0.5231
Humidity	0.0240
Light	2.8048
CO ₂	1.1943
Humidity ratio	-0.0545
Bias	-2.8849

Table 1: Feature Weights and Bias.

4 Multilayer perceptron

The second model implemented is a multilayer perceptron model (MLP), a feedforward neural network consisting of fully connected layers. Given the problem structure, the input layer must contain five neurons (one for each feature) while the output layer must have two layers. The probability of a sample belonging class y_k is computed as:

$$P(y = y_k | \mathbf{x}) = \hat{p}_k = \frac{e^{x_k}}{\sum_{i=0}^1 e^{x_i}} \quad (7)$$

The network parameters (weights and biases) are optimized by minimizing the cross-entropy loss function (6) using stochastic gradient descent. The good performance obtained with the logistic regression model suggests that the problem is well described by a low number of features and that a network with a complex architecture would only lead to overfitting without yielding meaningful performance gains. To test this hypothesis, three lightweight MLP architectures are evaluated:

- **Net 1:** One hidden layer with 2 neurons.
- **Net 2:** One hidden layer with 4 neurons.
- **Net 3:** Two hidden layers with 2 neurons each.

The validation set performance of these architectures was compared to select the optimal model.

Table 2 summarizes the structure (number of neurons for each layer), total number of parameters and accuracy on training, validation and test set of each one of the networks. These results confirm that the problem can be accurately described with a simple model and that increasing complexity

Network	Layers	Parameters	Train Acc (%)	Val Acc (%)	Test Acc (%)
Net 1	5 + 2 + 2	18	98.63	99.00	98.75
Net 2	5 + 4 + 2	34	98.65	99.25	98.75
Net 3	5 + 2 + 2 + 2	24	98.82	99.00	99.12

Table 2: Structure and accuracies of the MLP models.

does not yield significant improvements in the performance. Because of this, only the first network is kept for the following parts of the project.

Table 3 compares the performance obtained by the logistic regression model and the multilayer perceptron model (Net 1) on training and test set.

Model	Training accuracy (%)	Test accuracy (%)
Logistic regression	98.78	99.12
Multilayer perceptron	98.63	98.75

Table 3: Training and test accuracies of the models implemented in the project.

5 Error analysis

In order to better understand the behavior of the implemented models, their performance on the misclassified samples is analyzed. Figure 1 displays the confusion matrices for both the logistic regression model and the multilayer MLP. The logistic regression model exhibits a slight bias toward false positives (misclassifying unoccupied rooms as occupied), whereas the MLP demonstrates a more balanced error distribution, including false negatives (misclassifying occupied rooms as empty).

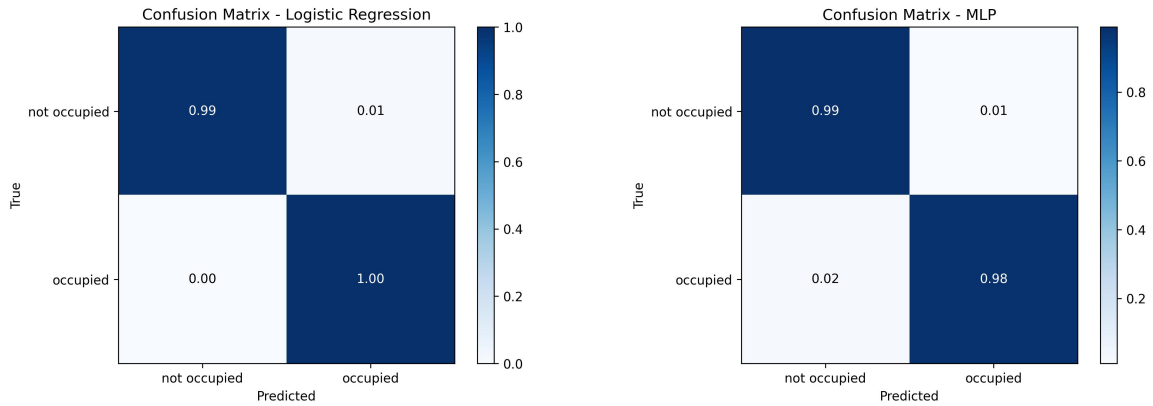


Figure 1: Confusion matrices corresponding to the logistic regression model (left) and the MLP (right).

Tables 4 and 5 report the normalized feature values of the samples misclassified by the logistic regression model and the MLP respectively along with the corresponding confidences. As shown in Table 4, the samples misclassified by logistic regression with the highest confidence (samples 433, 745, and 393) all exhibit high light intensity, further confirming that the model relies heavily on light for its decisions. Similarly, Table 5 indicates that the MLP misclassifies these exact same high-confidence false positive instances, demonstrating shared reliance on light intensity across both architectures. Additionally it shows that rooms with high temperatures are classified as empty. This behavior is consistent with the negative temperature weighting observed in the logistic regression caused by feature correlation.

Sample	True	Pred.	p	Temp.	Hum.	Light	CO ₂	Hum. Ratio
49	Empty	Occupied	0.8434	1.5777	0.0187	1.7036	0.5378	0.4907
198	Empty	Occupied	0.7260	-0.4435	-1.2684	1.4838	-0.4825	-1.3310
376	Empty	Occupied	0.7468	-0.4435	-1.2770	1.5426	-0.5309	-1.3392
393	Empty	Occupied	0.9706	2.0072	0.1534	2.0613	1.4131	0.7766
433	Empty	Occupied	0.9958	2.1941	0.1093	2.8381	1.3234	0.7890
655	Empty	Occupied	0.7356	-0.4189	-1.2724	1.5247	-0.5270	-1.3299
745	Empty	Occupied	0.9796	2.2268	0.0951	2.2823	1.3084	0.7838

Table 4: Normalized features and predicted probabilities for misclassified samples by the logistic regression model.

Sample	True	Pred.	p	Temp.	Hum.	Light	CO ₂	Hum. Ratio
49	Empty	Occupied	0.7376	1.5777	0.0187	1.7036	0.5378	0.4907
59	Occupied	Empty	0.8065	2.5121	0.2813	1.5605	0.3597	1.0922
198	Empty	Occupied	0.8505	-0.4435	-1.2684	1.4838	-0.4825	-1.3310
208	Occupied	Empty	0.8109	2.4826	0.2682	1.5605	0.3185	1.0670
376	Empty	Occupied	0.8675	-0.4435	-1.2770	1.5426	-0.5309	-1.3392
393	Empty	Occupied	0.9676	2.0072	0.1534	2.0613	1.4131	0.7766
433	Empty	Occupied	0.9975	2.1941	0.1093	2.8381	1.3234	0.7890
655	Empty	Occupied	0.8554	-0.4189	-1.2724	1.5247	-0.5270	-1.3299
745	Empty	Occupied	0.9768	2.2268	0.0951	2.2823	1.3084	0.7838
764	Occupied	Empty	0.8232	2.3350	0.2546	1.5222	0.2303	1.0006

Table 5: Normalized features and predicted probabilities for misclassified samples by the MLP.

6 Special request: precision and recall-oriented models

Apart from the test accuracy, other quantities can be defined to assess the performance of a model: the precision is defined as the fraction of positively classified samples that are actually positive:

$$P = \frac{TP}{TP + FP}, \quad (8)$$

while the recall is defined as the fraction of positive samples that are classified as positive by the model:

$$R = \frac{TP}{TP + FN}. \quad (9)$$

By choosing a higher confidence threshold, the number of false positives can be reduced, increasing the precision; conversely, a lower threshold can reduce the number of false negatives, increasing the recall. This means that there is always a trade-off between precision and recall, and they cannot be improved simultaneously.

By carefully tuning the bias for the logistic regression and the bias of one of the two output neurons for the MLP (which is equivalent to changing the confidence threshold), different versions of the two models were developed. Tables 6 and 7 show the values of validation accuracy, precision, and recall corresponding to different values of the bias for the logistic regression and the MLP respectively.

b	Acc.	Prec.	Rec.
-11	79.2%	100.0%	0.0%
-9	79.2%	100.0%	0.0%
-7	79.2%	100.0%	0.0%
-5	82.5%	100.0%	15.7%
-3	88.0%	97.3%	43.4%
-1	98.5%	97.5%	95.2%
1	98.5%	94.3%	98.8%
3	93.0%	74.8%	100.0%
5	45.0%	27.4%	100.0%
7	20.8%	20.8%	100.0%

Table 6: Precision and recall for the logistic regression.

b	Acc.	Prec.	Rec.
-11	79.2%	100.0%	0.0%
-9	79.2%	100.0%	0.0%
-7	82.0%	100.0%	13.3%
-5	84.8%	100.0%	26.5%
-3	91.5%	98.0%	60.2%
-1	98.2%	97.5%	94.0%
1	99.0%	96.5%	98.8%
3	97.0%	88.2%	98.8%
5	92.2%	72.8%	100.0%
7	20.8%	20.8%	100.0%

Table 7: Precision and recall for the multilayer perceptron.

Figure 2 shows the precision-recall curves for the two models.

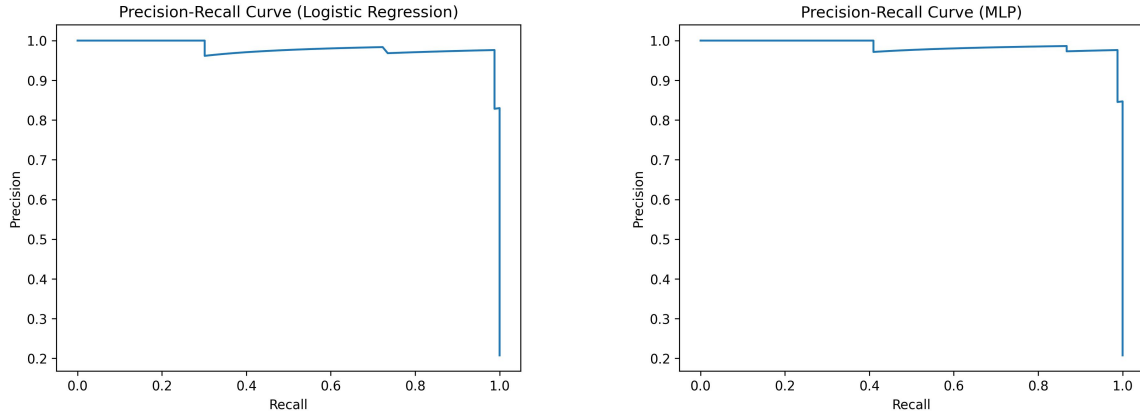


Figure 2: Precision-recall curves corresponding to the logistic regression model (left) and the MLP (right).

7 Conclusion

This project addressed the task of occupancy detection using machine learning: a logistic regression model and a multilayer perceptron model were trained to detect the presence of humans in a room based on environmental features and their performance was then assessed on a test set. Both models achieved a performance close to 99% on the test set. To better understand these models, the feature values of the misclassified samples were analyzed. Finally, by tuning the confidence threshold, precision-oriented and recall-oriented versions of the models were developed.

Use of AI

AI tools were used to improve the quality of the plots, fix the grammar and format the layout.

Integrity statement

I affirm that this report is the result of my own work and that I did not share any part of it with anyone else except the teacher.